

Trigonometry

Timothy Yau

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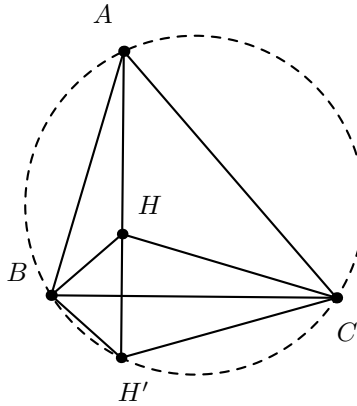
1 Introduction: Limitation of Simple Angle Chasing

We heard the phrases angle chasing and length chasing all the time, but what exactly are they?

Definition 1 Chasing refers to the process in which we express an unknown quantity in terms of some known quantity.

That is, we start with some fundamental quantities. For example, the interior angles of a particular triangle, say $\triangle ABC$. Then we express other quantities in the figure in terms of $\angle ABC$, $\angle BCA$ and $\angle CAB$. For example, if we would like to do the following problem:

Example 1 In $\triangle ABC$, H' is the reflection of the orthocenter H with respect to BC , show that H', A, B, C are concyclic.



[Solution]

Here, the basic quantities are $A = \angle BAC$, $B = \angle CBA$ and $C = \angle ACB$. Now we try to compute other angles, first notice that $BH \perp AC$, so by angle sum of triangle,

$$\angle HBC = 90^\circ - C$$

similarly

$$\angle HCB = 90^\circ - B$$

Therefore, again by angle sum of triangle,

$$\angle BHC = 180^\circ - (90^\circ - C) - (90^\circ - B) = B + C$$

Since H and H' are symmetric w.r.t. BC , we have

$$\angle BH'C = \angle BHC = B + C$$

Now it to verify that H', A, B, C are indeed concyclic, it suffices to prove

$$\angle BH'C + \angle BAC = 180^\circ$$

By substituting $\angle BH'C = B + C$ and $\angle BAC = A$, it suffices to show

$$A + B + C = 180^\circ$$

which is obvious. ■

Analysis. Let us review the solution. Imagine a paper writing down all the angles we know (that is, they can be expressed in terms of the fundamental angles A, B, C).

Initially this paper contains only A, B and C themselves. Next, we can express $\angle HBC, \angle HCB$ in terms of A, B, C , so these items are added to our paper. Similarly, $\angle BHC$ and $\angle BH'C$ are added to our paper in order. Notice two things:

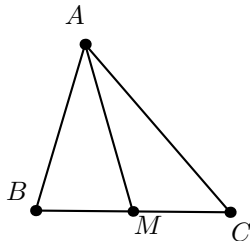
- We need tools for angle chasing, that is, they allow us to express angles in terms of other (more fundamental) angles. Without these tools, we cannot add anything to our paper, and angle chasing simply won't work! In simple angle chasing, there are 3 general tools serving this purpose:
 - Angle sum of triangle, which allows us to express an interior angle of a triangle in terms of the other two angles.
 - Similar triangles, which allow us to express an interior angle of a triangle in terms of the interior angle of another triangle.
 - Concyclic points, which allow us to obtain five(!) more equalities if we have established one.

all other elementary techniques you learnt are probably special cases of these three (eg. parallel lines is a special case of angle sum of triangle).

- Once we obtain our desired expression, we can transform our problem into proving an identity (similar to that in co-geom). This part is supposed to be trivial. (In the above example, this step is to show $A + B + C = 180^\circ$.)

Therefore, we will like to focus on the tool part. While angle sum of triangle, similar triangles and concyclic points are extremely useful, they still have a severe limitation: They are linear! So given some existing angles in our imaginary paper, what can be obtained are no other than the linear combination of the existing angles.

However, this is not always the case, consider the following simple configuration:



Here M is the midpoint of BC , can you express $\angle BAM, \angle CAM$ in terms of A, B and C ?

2 Trigonometric Theorems

I assume that you are familiar with the basic trigonometric functions.

2.1 Very Useful Theorems

Theorem 0 These theorems have important geometrical meaning

$$\sin(180^\circ - x) = \sin x$$

$$\cos(180^\circ - x) = -\cos x$$

$$\sin(90^\circ - x) = \cos x$$

$$\cos(90^\circ - x) = \sin x$$

Theorem 1 (Sine Law) In $\triangle ABC$, we have

$$\frac{\sin \angle ABC}{AC} = \frac{\sin \angle BCA}{BA} = \frac{\sin \angle CAB}{CB} = 2R$$

where R is the circumradius.

Theorem 2 (Double angle Formulae) For any $x \in \mathbb{R}$,

$$\sin 2x = 2 \sin x \cos x$$

$$\cos 2x = 2 \cos^2 x - 1 = 1 - 2 \sin^2 x = \cos^2 x - \sin^2 x$$

$$\tan 2x = \frac{2 \tan x}{1 - \tan^2 x}$$

Theorem 3 (Product-to-sum Formulae) For any $x, y \in \mathbb{R}$,

$$\sin(x + y) + \sin(x - y) = 2 \sin x \cos y$$

$$\sin(x + y) - \sin(x - y) = 2 \cos x \sin y$$

$$\cos(x + y) + \cos(x - y) = 2 \cos x \cos y$$

$$\cos(x + y) - \cos(x - y) = -2 \sin x \sin y$$

Theorem 4 (sum-to-product Formulae)

$$\sin x + \sin y = 2 \sin \frac{x + y}{2} \cos \frac{x - y}{2}$$

$$\sin x - \sin y = 2 \cos \frac{x + y}{2} \sin \frac{x - y}{2}$$

$$\cos x + \cos y = 2 \cos \frac{x + y}{2} \cos \frac{x - y}{2}$$

$$\cos x - \cos y = -2 \sin \frac{x + y}{2} \sin \frac{x - y}{2}$$

2.2 Less Useful Theorems

Theorem 5 (Cosine Law) Suppose $\triangle ABC$ is a triangle then

$$BC^2 = AB^2 + AC^2 - 2 \cdot AB \cdot AC \cdot \cos \angle BAC$$

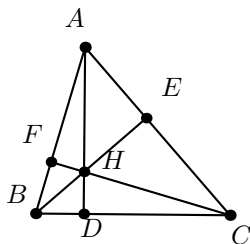
Theorem 6 (Compound angle formulae) Suppose $x, y \in \mathbb{R}$ then

$$\sin(x + y) = \sin x \cos y + \sin y \cos x$$

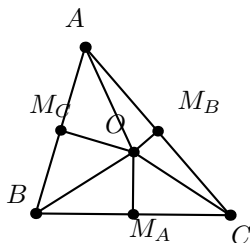
$$\cos(x + y) = \cos x \cos y - \sin x \sin y$$

3 Application of sine law: Computing length in triangle

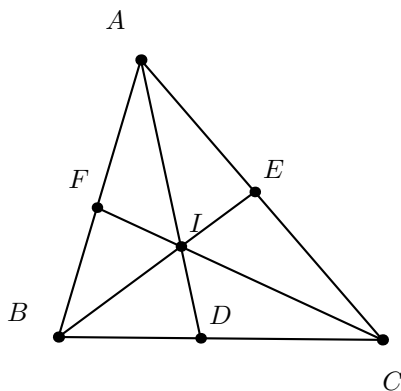
Using the (extended) sine law, we can compute a lot of length in the triangle. In the following examples, express the desired length in terms of R, A, B, C (possibly with trigonometric functions)



- (i) AB, BC, CA
- (ii) AD, BE, CF
- (iii) AH, BH, CH
- (iv) BD, CD, CE, EA, AF, FB



- (v) AO, BO, CO
- (vi) OM_A, OM_B, OM_C

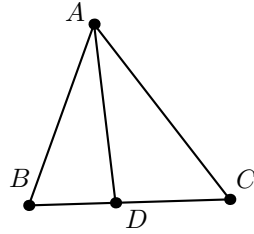


- (vii) AD, BE, CF
- (viii) AI, BI, CI
- (ix) ID, IE, IF
- (x) r , the inradius of $\triangle ABC$
- (xi) BD, CD, CE, EA, AF, FB

4 Application of sine law: Ratio Lemma

Theorem 7 Given $\triangle ABC$, suppose D is a point on BC then

$$\frac{BD}{DC} = \frac{\sin \angle BAD}{\sin \angle CAD} \cdot \frac{BA}{CA}$$



[Solution]

There are two solutions:

[Solution 1]

$$\frac{BD}{DC} = \frac{[BAD]}{[CAD]} = \frac{\frac{1}{2}AB \cdot AD \sin \angle BAD}{\frac{1}{2}AD \cdot AC \sin \angle CAD} = \frac{\sin \angle BAD}{\sin \angle CAD} \cdot \frac{BA}{CA}$$

[Solution 2]

$$\frac{BD}{DC} = \frac{BD}{DA} \cdot \frac{DA}{CD} = \frac{\sin \angle BAD}{\sin \angle ABD} \cdot \frac{\sin \angle CAD}{\sin \angle ACD} = \frac{\sin \angle BAD}{\sin \angle CAD} \cdot \frac{\sin \angle ACB}{\sin \angle ABC} = \frac{\sin \angle BAD}{\sin \angle CAD} \cdot \frac{BA}{CA}$$

■

While Solution 1 is short and elegant, Solution 2 shows an important idea. Though we cannot directly compare BD and DC , we can introduce an intermediate quantity DA , we can compute the ratio BD/DA and DA/CD using sine law, and the result easily follows.

Example 2 Using Theorem 7,

- Deduce the angle bisector theorem (both the internal and external ones)
- Show that if D is the midpoint of BC then

$$\frac{CA}{BA} = \frac{\sin \angle BAD}{\sin \angle CAD}$$

Corollary 1 Suppose D, E are two points in the interior of $\angle BAC$, show that A, D, E are collinear if and only if

$$\frac{\sin \angle BAD}{\sin \angle CAD} = \frac{\sin \angle BAE}{\sin \angle EAD}$$

[Remark.] This corollary is very useful. However, it needs to be used carefully, you need to ensure that the two points belong simultaneously to the interior of $\angle BAC$ first, which usually involves some tedious case issues.

5 Application of sine law: Ceva and Menelaus Theorem

Theorem 8 (Trigonometric form of Ceva's Theorem) Cevians AD, BE, CF concur if and only if

$$\frac{\sin \angle BAD}{\sin \angle DAC} \cdot \frac{\sin \angle CBE}{\sin \angle EBA} \cdot \frac{\sin \angle ACF}{\sin \angle FCB} = 1$$

Example 3 Try to draw three different configurations of the Ceva's Theorem.

Theorem 9 (Menelaus Theorem) If line PQR intersecting AB on $\triangle ABC$, where P is on BC , Q is on the extension of AC , and R on the intersection of PQ and AB , then

$$\frac{PB}{CP} \cdot \frac{QC}{QA} \cdot \frac{AR}{RB} = 1.$$

Example 4 Try to draw two different configurations of the Menelaus' Theorem.

[Remark.] Ceva and Menelaus are in my opinion the most underrated theorems in IMO. Some contestants has never even heard about the trigonometric form of Ceva, let alone using it. There are some ways to apply it

- Use them to show collinearity and concurrency (most people know this)
- Utilizing the symmetry of Ceva and Menelaus.
- Use Theorem 8 and 9 together with Theorem 7 or corollary 1 to obtain a lot of information regarding cevians.
- The theorem is very general.
- It can be used to reduce the number of points.

6 More Advanced Tools

Theorem 10 (Circle Ratio Lemma) Suppose A, B, C, D are concyclic. Suppose AD and BC intersect at E , then

$$\frac{BE}{EC} = \frac{AB}{AC} \cdot \frac{DB}{DC}$$

for more advanced applications, see <https://cdn.artofproblemsolving.com/attachments/2/b/cfc6469f8aa90eee54549e421c5080108bd189.pdf>

Projective Geometry Projective geometry is a deep topic, however, its application in Olympiad is essentially trigonometry. Most of the fundamental theorems, such as the invariance of cross-ratio under perspectivity, Pascal's theorem, Pappus Theorem and the properties of harmonic bundles can all be shown using the theorems covered above.

These tools (along with Theorem 10) can be seen as short-cuts when applying trigonometry. It is analogous to the case in which the derivative of all elementary functions can be computed using first principle. However, we can simplify our work a lot by using more sophisticated methods such as product rule, chain rule, logarithmic differentiation, etc.

7 Exercises

7.1 Easy

1. (Korea MO Final 2011 P5) ABC is a triangle such that $AC < AB < BC$ and D is a point on side AB satisfying $AC = AD$. The circumcircle of ABC meets with the bisector of angle A again at E and meets with CD again at F . K is an intersection point of BC and DE . Prove that $CK = AC$ is a necessary and sufficient condition for $DK \cdot EF = AC \cdot DF$.
2. (IOM 2021 P2) Points P and Q are chosen on the side BC of triangle ABC so that P lies between B and Q . The rays AP and AQ divide the angle BAC into three equal parts. It is known that the triangle APQ is acute-angled. Denote by B_1, P_1, Q_1, C_1 the projections of points B, P, Q, C onto the lines AP, AQ, AP, AQ , respectively. Prove that lines B_1P_1 and C_1Q_1 meet on line BC .
3. (China TST 2012 Test 1 P2) Given a scalene triangle ABC . Its incircle touches BC, AC, AB at D, E, F respectively. Let L, M, N be the symmetric points of D with EF , of E with FD , of F with DE , respectively. Line AL intersects BC at P , line BM intersects CA at Q , line CN intersects AB at R . Prove that P, Q, R are collinear.
4. (Geometry in Figures 4.3.7) In $\triangle ABC$, the angle bisectors of $\angle ABC$ intersect AC at D and the circumcircle of $\triangle ABC$ again at M . The angle bisectors of $\angle ACB$ intersect AB at E and the circumcircle of $\triangle ABC$ again at N . Suppose MN and ED intersect at X , show that XA is tangent to the circumcircle of $\triangle ABC$.
5. (Geometry in Figures 4.3.16) In $\triangle ABC$, the external angle bisector of $\angle BAC$ meets the side BC at Q . Let P, Q be the foot of perpendicular from B to AD , suppose E is the intersection of BQ and AC while F is the intersection of CP and AB . Show that D, F, E are collinear.

7.2 Medium

1. (HKIMO TST 2021 Test 2 P3) Let $\triangle ABC$ be an acute triangle with circumcircle Γ , and let P be the midpoint of the minor arc BC of Γ . Let AP meet BC at D , and let M be the midpoint of AB . Also, let E be the point such that $AE \perp AB$ and $BE \perp MP$. Prove that $AE = DE$.
2. (All-Russian MO 2012 Grade 10 P8) The point E is the midpoint of the segment connecting the orthocentre of the scalene triangle ABC and the point A . The incircle of triangle ABC is tangent to AB and AC at points C' and B' respectively. Prove that point F , the point symmetric to point E with respect to line $B'C'$, lies on the line that passes through both the circumcentre and the incentre of triangle ABC .
3. (IOM 2020 P6) In convex pentagon $ABCDE$ points A_1, B_1, C_1, D_1, E_1 are intersections of pairs of diagonals (BD, CE) , (CE, DA) , (DA, EB) , (EB, AC) and (AC, BD) respectively. Prove that if four of quadrilaterals AB_1A_1B , BC_1B_1C , CD_1C_1D , DE_1D_1E and EA_1E_1A are cyclic then the fifth one is also cyclic.
4. (USATST 2019 P5) Let ABC be an acute triangle with orthocenter H and circumcircle Γ . A line through H intersects segments AB and AC at E and F , respectively. Let K be the circumcenter of $\triangle AEF$, and suppose line AK intersects Γ again at a point D . Prove that line HK and the line through D perpendicular to $\overline{BC}$ meet on Γ .
5. (CMO 2019 P2) In triangle ABC , $AB > AC$. The bisector of $\angle BAC$ meets BC at D . P is on line DA , such that A lies between P and D . PQ is tangent to $\odot(ABD)$ at Q . PR is tangent to $\odot(ACD)$ at R . CQ meets BR at K . The line parallel to BC and passing through K meets QD, AD, RD at E, L, F , respectively. Prove that $EL = KF$.
6. (IMO Shortlist 2021 G4) Let $ABCD$ be a quadrilateral inscribed in a circle Ω . Let the tangent to Ω at D meet rays BA and BC at E and F , respectively. A point T is chosen inside $\triangle ABC$ so that $\overline{TE} \parallel \overline{CD}$ and $\overline{TF} \parallel \overline{AD}$. Let $K \neq D$ be a point on segment DF satisfying $TD = TK$. Prove that lines AC, DT , and BK are concurrent.

7.3 Hard

1. (IMO shortlist 2014 G6) Let ABC be a fixed acute-angled triangle. Consider some points E and F lying on the sides AC and AB , respectively, and let M be the midpoint of EF . Let the perpendicular bisector of EF intersect the line BC at K , and let the perpendicular bisector of MK intersect the lines AC and AB at S and T , respectively. We call the pair (E, F) *interesting*, if the quadrilateral $KSAT$ is cyclic. Suppose that the pairs (E_1, F_1) and (E_2, F_2) are interesting. Prove that $\frac{E_1E_2}{AB} = \frac{F_1F_2}{AC}$.
2. (IMO Shortlist 2015 G7) Let $ABCD$ be a convex quadrilateral, and let P, Q, R , and S be points on the sides AB, BC, CD , and DA , respectively. Let the line segment PR and QS meet at O . Suppose that each of the quadrilaterals $APOS, BQOP, CROQ$, and $DSOR$ has an incircle. Prove that the lines AC, PQ , and RS are either concurrent or parallel to each other.

3. (IMO Shortlist 2016 G6) Let $ABCD$ be a convex quadrilateral with $\angle ABC = \angle ADC < 90^\circ$. The internal angle bisectors of $\angle ABC$ and $\angle ADC$ meet AC at E and F respectively, and meet each other at point P . Let M be the midpoint of AC and let ω be the circumcircle of triangle BPD . Segments BM and DM intersect ω again at X and Y respectively. Denote by Q the intersection point of lines XE and YF . Prove that $PQ \perp AC$.
4. (IMO shortlist 2019 G6) Let I be the incentre of acute-angled triangle ABC . Let the incircle meet BC, CA , and AB at D, E , and F , respectively. Let line EF intersect the circumcircle of the triangle at P and Q , such that F lies between E and P . Prove that $\angle DPA + \angle AQD = \angle QIP$.
5. (USAMO 2016 P5) An equilateral pentagon $AMNPQ$ is inscribed in triangle ABC such that $M \in \overline{AB}$, $Q \in \overline{AC}$, and $N, P \in \overline{BC}$. Let S be the intersection of $\overleftrightarrow{MN}$ and $\overleftrightarrow{PQ}$. Denote by ℓ the angle bisector of $\angle MSQ$.
Prove that $\overline{OI}$ is parallel to ℓ , where O is the circumcenter of triangle ABC , and I is the incenter of triangle ABC .

7.4 The Ultimate Challenge

1. (IMO shortlist 2021 G8) Let ABC be a triangle with circumcircle ω and let Ω_A be the A -excircle. Let X and Y be the intersection points of ω and Ω_A . Let P and Q be the projections of A onto the tangent lines to Ω_A at X and Y respectively. The tangent line at P to the circumcircle of the triangle APX intersects the tangent line at Q to the circumcircle of the triangle AQY at a point R . Prove that $\overline{AR} \perp \overline{BC}$.